

REQUIREMENT ANALYSIS DOCUMENT (RAD)

Project Title: Online Shopping Database Management System (Implementation: SmartShop)

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1. Executive Project Summary

SmartShop is an online shopping platform that enables customers to register, browse products, add items to a shopping cart, place orders, make secure online payments, and track shipments. The system provides a user-friendly shopping experience while allowing administrators to manage products, customers, orders, inventory, and deliveries efficiently.

To ensure secure and reliable data management, SmartShop uses a Relational Database Management System (RDBMS) such as MySQL or PostgreSQL. A relational database maintains data accuracy, prevents duplicate records, and supports multiple users performing transactions simultaneously.

The system follows ACID properties to ensure reliable transactions:

- **Atomicity:** When a customer places an order, both payment confirmation and order creation must complete successfully. If payment fails, the entire transaction is cancelled.
 - **Consistency:** The database ensures that invalid data cannot be stored. Product prices cannot be negative, and customers cannot purchase products that are out of stock.
 - **Isolation:** Multiple customers can place orders at the same time without affecting each other's transactions.
 - **Durability:** Once an order is successfully completed, it remains permanently stored in the database even if the system experiences a failure.
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2. Core Business Domains & Features

2.1 Customer Management

This module manages customer registration, login credentials, profile information, and shipping addresses.

Business Rules

- Every customer must register using a unique email address.
- Each customer is assigned a unique Customer ID.
- Customers can update their personal information and delivery addresses.
- Customers can browse products before making a purchase.
- A customer may have multiple orders in the system.

2.2 Product & Inventory Management

This module stores product details, categories, pricing, and stock information.

Business Rules

- Every product has a unique Product ID.
- Each product belongs to one category.
- Product prices must always be greater than zero.
- Stock quantity must be updated immediately after every successful purchase.
- Products that are out of stock cannot be ordered.

2.3 Order Management

This module records customer purchases and manages the order lifecycle.

Business Rules

- Every order has a unique Order ID.
- One customer can place multiple orders.
- One order can contain multiple products.
- Order status follows the sequence:
 - Pending
 - Confirmed
 - Shipped
 - Delivered
 - Cancelled
- Customers can view their complete order history.

2.4 Payment & Shipment Management

This module manages online payments and product delivery.

Business Rules

- Every order must have one payment record.
- Payment must be successful before shipment begins.
- Every shipment receives a unique Tracking ID.
- Customers can track the shipment status at any time.

- Delivery status is updated until the order reaches the customer.

3. Actor Profiles & System Interactions

1. Customer

What they can Read

- Product details
- Product categories
- Their own order history
- Shipment tracking information

What they can Write/Edit

- Register a new account
- Update profile information
- Place orders
- Make payments
- Submit product reviews

Safety Boundary

Customers cannot access or modify another customer's personal information, payment details, or order history.

2. Store Manager

What they can Read

- Product inventory
- Customer orders
- Sales reports

What they can Write/Edit

- Add new products
- Update product prices
- Manage inventory
- Process customer orders

Safety Boundary

Store managers cannot view customer passwords or payment information.

3. Delivery Staff

What they can Read

- Customer delivery address
- Assigned delivery orders
- Shipment details

What they can Write/Edit

- Update shipment status
- Mark orders as delivered

Safety Boundary

Delivery staff cannot modify product details, prices, or customer payment records.

4. System Administrator

What they can Read/Write

- Full access to all database tables and system information.
- Create, update, delete, and manage all records.

Safety Boundary

All administrator activities are recorded in audit logs for security purposes.

4. Analytical Reporting Requirements

Report 1 – Low Stock Products Report

Business Value

Helps the store manager identify products that require immediate restocking.

Report Includes

- Product Name
 - Category
 - Available Stock Quantity
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Report 2 – Top Customers Report

Business Value

Identifies loyal customers based on their total purchases.

Report Includes

- Customer Name
 - Email Address
 - Total Orders
 - Total Amount Spent
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Report 3 – Monthly Sales Report

Business Value

Helps management analyze monthly business performance and product sales.

Report Includes

- Product Category
 - Number of Orders
 - Total Revenue
 - Best Selling Products
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5. Non-Functional Requirements & Database Constraints

Data Constraints

- Customer email addresses must be unique.
 - Product prices must always be greater than zero.
 - Stock quantity cannot be negative.
 - Payment amount must match the total order value.
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Concurrency Control

If two customers attempt to purchase the last available product at the same time, the database locks the inventory record. The first successful transaction reserves the product, while the second customer is notified that the product is out of stock.

Security Requirements

- Customer passwords must be encrypted.
- Payment information must be securely stored.
- Only authorized users can access sensitive system data.
- All user activities should be logged for security purposes.

Historical Records & Safe Deletion

Customer accounts should not be permanently deleted from the database. Instead, the account status should be changed to **Inactive**. Previous orders and payment records must remain available for future reporting, auditing, and business analysis.

Conclusion

The SmartShop Online Shopping Database Management System is designed to provide a secure, reliable, and efficient platform for online shopping. The Requirement Analysis Document identifies the major business processes, user roles, business rules, reporting requirements, and database constraints. This document serves as the foundation for designing the database tables, relationships, and SQL implementation in the upcoming weeks.